

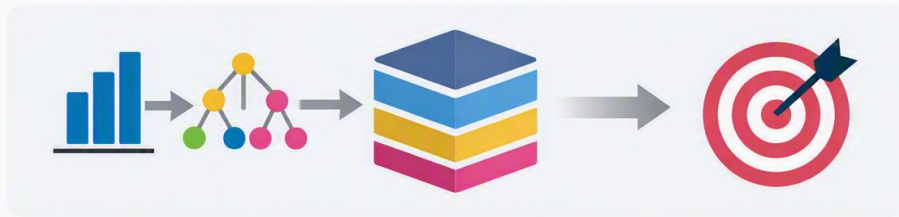
Part A: Ensemble Learning Answers

1. What are Ensemble Learning Methods and why are effective?



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What are Ensemble Learning Methods and why are they *effective*?



- Combines multiple models to form a stronger predictor.
- Improves **accuracy** and reduces **overfitting**.
- Enhances generalization by leveraging *diverse model strengths*.

Ensemble Learning Methods involve combining multiple base models to create a stronger and more robust model. They are effective because they reduce overfitting, improve accuracy, and provide better generalization by leveraging the strengths of diverse models.

2. Explain the difference between Bagging, Boosting, and Voting.

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Bagging vs. Boosting vs. Voting



Bagging: Trains models independently on random subsets to reduce variance.



Boosting: Sequentially corrects errors of previous models to reduce bias.



Voting: Combines predictions via majority or weighted votes for better accuracy.

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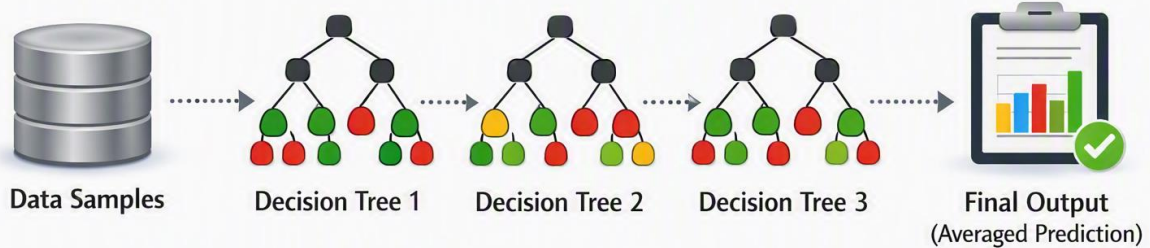
3. What is Random Forest and how does it work?

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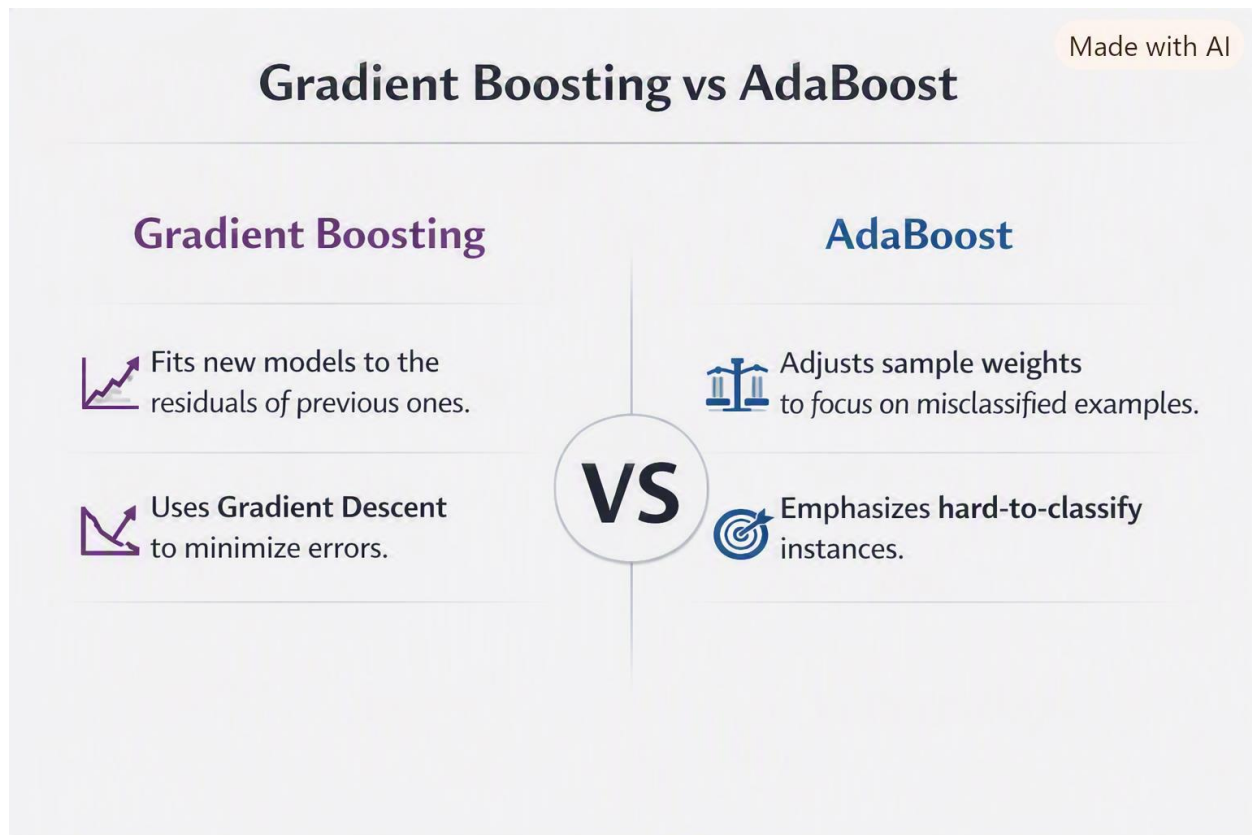
What is Random Forest and how does it work?

Random Forest is an ensemble of decision trees built on random subsets of data and features. It averages their outputs to improve stability and reduce overfitting.



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4. What is Gradient Boosting and how is it different from AdaBoost?



Gradient Boosting minimizes errors by fitting new models to the residuals of previous ones using gradient descent. AdaBoost adjusts sample weights instead, emphasizing misclassified examples.

5. Why is ensemble learning preferred over a single model?



Why is Ensemble Learning Preferred?

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- ✓ Captures multiple perspectives.
- ✓ Reduces both bias and variance.
- ✓ Delivers **more reliable** predictions.
- ✓ Outperforms individual models in **real-world tasks**.



It captures multiple perspectives, reduces both bias and variance, and delivers more reliable predictions — outperforming individual models in most real-world tasks.